Data Model and DAX Measures

1. Overview

This document describes the data model, relationships, business measures, and calculation logic used in the Adventure Works Business Intelligence project.
The report analyses sales performance across products, geography, sales channels, and resellers using Power BI Desktop and DAX.

2. Data Model

The model connects transactional sales data with descriptive tables to support analysis across multiple business dimensions.

Table	Purpose
Sales	Transaction-level sales data and sales-related keys
Product	Product names, models, categories, and subcategories
Reseller	Reseller names, business types, and reseller attributes
Date / Calendar	Dates, months, quarters, and years
Geography / Territory	Countries, regions, and sales territories
Customer	Customer attributes, if available in the model
Channel	Sales channel information, if available

3. Relationships

Document the relationships between the tables, including:

- The primary key and foreign key used.
- Relationship cardinality, such as one-to-many.
- Cross-filter direction.
- Whether the relationship is active or inactive.
- How the relationship affects report filtering.

Reseller relationship

The Reseller table is connected to the Sales table through the reseller key.
This relationship enables reseller-level sales analysis without duplicating reseller information in every report visual.
Rows with a blank reseller key should be interpreted according to the source data and sales channel logic. A populated reseller key alone should not be assumed to prove that a transaction belongs to the reseller channel without validating the model.

4. DAX Measures

Total Sales

Purpose: Calculates sales revenue within the selected filter context.

```
Total Sales =
SUM(Sales[SalesAmount])
```

Total Orders

Purpose: Counts distinct sales orders.

```
Total Orders =
DISTINCTCOUNT(Sales[SalesOrderNumber])
```

Total Profit

Purpose: Calculates profit using sales revenue and the applicable cost field.

```
Total Profit =
[Total Sales] - [Total Cost]
```

The definition of Total Cost must match the grain and meaning of the source cost data.

Profit Margin %

Purpose: Measures profit as a proportion of sales.

```
Profit Margin % =
DIVIDE(
  [Total Profit],
  [Total Sales],
  0
)
```

Format as a percentage.

Sales Previous Year

Purpose: Retrieves sales for the comparable prior-year period.

```
Sales Previous Year =
CALCULATE(
  [Total Sales],
  SAMEPERIODLASTYEAR('Date'[Full Date].[Date]
)
```

This measure requires a correctly configured Calendar table and appropriate date relationships.

Sales Growth %

Purpose: Measures the percentage change in sales against the previous year.

```
Sales Growth % =
DIVIDE(
  [Total Sales] - [Sales Previous Year],
  [Sales Previous Year]
)
```

Total Reseller Sales

Purpose: Calculates sales associated with records that have a reseller key, where this is consistent with the model's channel logic.

```
Total Reseller Sales =
CALCULATE(
  [Total Sales],
  Sales[ResellerKey] <> BLANK()
)
```

Validate this measure against the channel definition and total sales reconciliation before using it as the official reseller KPI.

5. Time Intelligence

The report includes historical performance comparisons such as:

- Sales versus the previous year.
- Monthly sales changes.
- Profit growth by region and category.
- Reseller sales growth.

All time-based measures should use a consistent Calendar table and documented rules for missing periods and incomplete data.

6. Filter Context

The report uses slicers and visual interactions to analyse performance by:

- Year
- Month
- Product category and model
- Country and region
- Reseller and business type
- Sales channel

7. Validation Checklist

- Total sales reconcile with the source data.
- Order counts use the correct order identifier.
- Reseller sales reconcile with the intended channel population.
- Profit and profit margin use consistent cost definitions.
- Previous-year measures return the correct comparison period.
- Missing prior-period values are not incorrectly presented as genuine declines.
- Totals are recalculated correctly rather than averaging row-level percentages.
- KPI targets are clearly identified as actual or illustrative.

8. Assumptions and Limitations

This project uses a sample dataset for demonstration and portfolio purposes. Any illustrative targets should be distinguished from actual business targets. Analytical findings should be interpreted within the dataset's date range, data quality, and business definitions.

9. Tools

- Power BI Desktop

- DAX
- Power Query
- Relational data modelling
- Interactive data visualization